

Flicker Noise Analysis on Chopper Amplifier

Ting Zhou, Zhuo Gao, Jiajie Huang, Yewangqing Lu, Mingyi Chen, Yongfu Li*
 Department of Micro-Nano Electronics and MoE Key Lab of Artificial Intelligence
 Shanghai Jiao Tong University, China
 *Email: yongfu.li@sjtu.edu.cn

Abstract—This paper aims to maximize the performance of the chopper amplifier, considering the design trade-off between noise and power consumption. To achieve the goal, we identify and model the various sources of flicker noise in the amplifier and switches using Verilog-AMS language. Then, we build a chopper amplifier circuit and investigate the influences of chopping frequency, the type of switches (*N*-type switch, *N*-type switch with dummy, transmission gate, and bootstrapped *N*-type switch) and its non-idealities on its noise performance. Based on our analytical study with simulation results using the circuit model, we can identify the optimal chopping frequency and use *N*-type switches to minimize mismatch in parasitic capacitance. Therefore, we can reduce the total input-referred-noise by $4.9\times$ compared with an amplifier without using the chopping technique.

Index Terms—Flicker Noise, Chopping Technique, Signal Modulation, Charge Injection, Clock Feed-through, Bootstrapped Switch

I. INTRODUCTION

Flicker noise is an electronic noise with a $1/f$ power spectral density ($1/f$ noise), causing huge interference to low frequency signals [1]. Often, it has limited the performance of precision measurement sensors [2]. Thus, it is pivotal to incorporate noise reduction peripheral circuit to subjugate the noise level in sensor circuits.

According to the international roadmap for devices and systems (IRDS), the down-scaling of CMOS technology has exacerbated the effect of $1/f$ noise [3]. Intuitively, increasing the transistors' gate area helps to minimize the $1/f$ noise. However, this method will significantly increase the area and parasitic capacitance, which is self-contradicting to CMOS scaling. Several noise reduction techniques, such as chopping technique [4], auto-zeroing technique [5], and correlated double sampling technique [6], have been used in operational amplifiers to reduce the noise by $1\sim 2$ order [7], [8]. Among these techniques, the chopping technique is a more preferable technique for its superior capability in suppressing flicker noise [9]. It is a continuous-time modulation technique that does not cause noise folding comparing to the auto-zeroing technique and correlated double sampling technique. This technique has widely been used in smart sensors, sensor interfaces, medical instruments, and precision voltage references [10]–[12].

There are several design challenges when using the chopping technique, where special attention must be paid to the circuit analysis, design, and implementation. The chopping frequency and its clock source play an important design

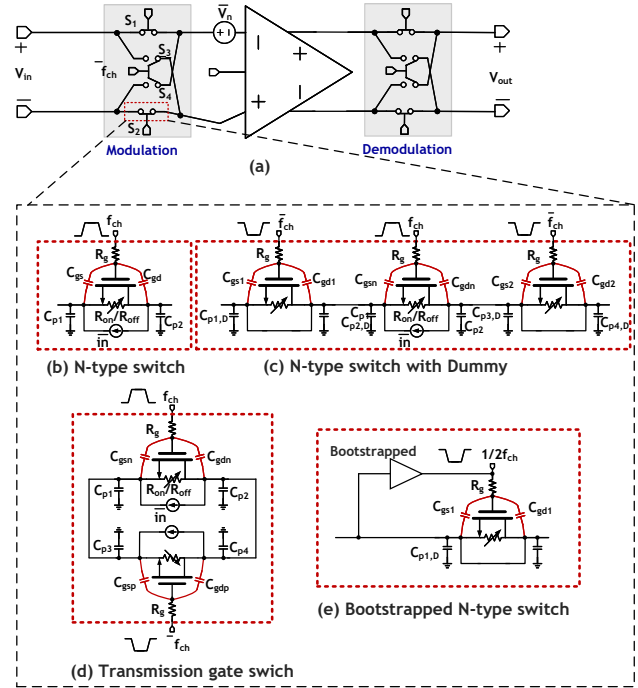


Fig. 1. Non-idealities and Different types of noise models in amplifier and switches (a) operational amplifier (b) *N*-type switch (NS), (c) *N*-type switch with dummy switch (NS-D), (d) transmission gate (TG) and (e) bootstrapped *N*-type switch (BS).

trade-off between the total power consumption and the amount of noise reduction. Ref. [13] has demonstrated the important of this study with measurement result; however, we believe that there is a need to provide an accurate circuit modelling, include circuit non-ideality and flicker noise model, to design a more complex study systematically and accurately.

In this work, we have developed analog switches and amplifiers behaviour circuit models using the Verilog-AMS programming language. These models include the non-ideal parasitic resistor and capacitor components and use the recently corrected flicker noise model [14]. These circuit models enabled us to design and analyze the noise level of a chopper amplifier under the different types of switches and different characteristics of the chopper clock.

II. ANALYZING AND MODELING OF FLICKER NOISE IN SWITCHES AND AMPLIFIER

A. Sources of Flicker Noise

Due to the random charges trapping and de-trapping process near the interface layer between the silicon and silicon

```

module amp_model(voutp, voutn, vref, vip, vin, vdd,
vss);...
analog function integer sign;
    input arg;
    real arg;
    sign = arg >= 0 ? +1 : -1;
endfunction

analog begin
...
@(initial_step or initial_step("dc")) begin
    Ir = V(vip, vin)/R;
    Pn = KF*pow(abs(Ir), AF);
end
I(vip, vin) <+ white_noise(4.0* P_K*$temperature/Rt, "thermal");
I(vip, vin) <+ flicker_noise(sign(Ir)*Pn, EF, "flicker");
vin_val = -V(vip, vin)/2 + vin_offset + rin * I(vip, vin);
...
end
endmodule

```

Fig. 2. A code snippet of the proposed Verilog-AMS behavioral model for amplifier

dioxide (transistor's gate), these fluctuations affect the channel carrier density, which in turn contributes to the flicker noise characteristics compared to the bipolar devices [15]. The power spectrum density (*PSD*) of flicker noise exhibits a $1/f$ spectral characteristic, where its equivalent flicker noise voltage source is expressed as

$$\overline{V_n^2} = \frac{K}{C_{ox}} WL \frac{1}{f}, \quad (1)$$

where K is process-dependent flicker noise constant, C_{ox} is the gate oxide capacitance per unit area, W and L are the effective gate width and length of the transistor, respectively. From the equation, $\frac{K}{C_{ox}} WL$ represents the frequency-dependent noise power at 1 Hz. In the Verilog-AMS model, there is a flicker noise function :

$$flicker_noise(pwr, EF, "flicker") \quad (2)$$

where pwr represents noise power at 1 Hz and EF represents the exponential factor of $1/f$.

B. Modeling of Non-idealities in Amplifier

The Verilog-AMS behavioral model for operational amplifier has been used in the system modeling of sigma-delta modular [16]. This operational amplifier is built with limited unit-gain bandwidth, slew-rate, and nonlinear gain. To further improve the accuracy of the circuit modeling, we have incorporated the recently corrected flicker noise model [14] and thermal noise into the amplifier model [16]. The code snippet of the proposed Verilog-AMS behavioral model for the operational amplifier is shown in Fig. 2. The amplifier is configured with a voltage gain of 20 dB and a bandwidth of 30 kHz, which is a common design specification for biomedical readout circuit [17]–[21].

C. Types of Switches (*NS*, *NS-D*, *TG*, and *BS*)

The design and uses of switches have become very critical in noise reduction techniques to subjugate the noise level in an operational amplifier. These switches suffer

```

module switch_model(in,
out);...
analog begin
    @(initial_step or initial_step("dc")) begin
        Ir = V(in, out)/R;
        Pn = KF*pow(abs(Ir), AF);
    end
    V(r) <+ 1/(K*Ratio*(V(C,in)-vth));
    I(in, out) <+ white_noise(4.0* P_K*$temperature/R, "thermal");
    I(in, out) <+ flicker_noise(sign(Ir)*Pn, EF, "flicker");
    if (V(C)>vth? 1:0)
        V(in, out)<+ I(in, out)/(K*Ratio*(V(C,in)-vth));
    else
        V(in, out)<+ I(in, out)*roff;
    end
endmodule

```

Fig. 3. Verilog-AMS Code snippet for modeling the signal-dependent on-resistance and noises in the switches

from four commonly-used circuit's non-idealities, namely (1) signal-dependent on-resistance, (2) channel charge injection through the source and drain terminals, (3) clock feedthrough: charge injection caused by overlap coupling between the gate and drain, and (4) thermal and flicker noises.

As presented in Fig. 1, there are four common types of switches used in the circuit design, namely *N*-type switch (*NS*), *N*-type switch with dummy (*NS-D*), transmission gate (*TG*), and bootstrapped *N*-type switch (*BS*) [13].

***N*-type switch (*NS*):** As shown in Fig. 1(b), *NS* is the simplest switch using a *N*-type transistor, however, it has the worst signal-dependent on-resistance among the switches, introducing signal distortion to its output.

***N*-type switch with dummy (*NS-D*):** As shown in Fig. 1(c), *NS-D* uses additional *N*-type transistors in series of the switch transistor, however, it is hard to keep the equal splitting of the charge between the dummy and switch transistors.

Transmission gate (*TG*): As shown in Fig. 1(d), *TG* is a bilateral switch consisting of a *N*-type and a *P*-type transistor, where the gate is connected to the complementary clock signal. It can achieve a constant on-resistance with limited dynamic range but sensitive to (1) the mismatch in *N*-type and *P*-type transistors, and (2) misalignment of clock's edges.

Bootstrapped *N*-type switch (*BS*): As shown in Fig. 1(e), *BS* eliminates the signal-dependent on-resistance at the expense of additional area and power consumption. It can be used with a dummy switch to eliminate the problem with charge injection.

D. Modeling of Non-idealities in Switches

The four commonly-used circuit's non-idealities can be illustrated in Fig. 1(b)–(e). As shown in Fig. 3, we proposed to use Verilog-AMS to model the (1) signal-dependent on-resistance and (2) thermal and flicker noises effects in these switches. To model the channel charge injection and clock feedthrough, we have added parasitic capacitors C_{gs} , C_{gd} , C_{p1} and C_{p2} to these switches and the values are dependent on the transistor's gate dimension.

III. UNDERSTANDING CHOPPER AMPLIFIER

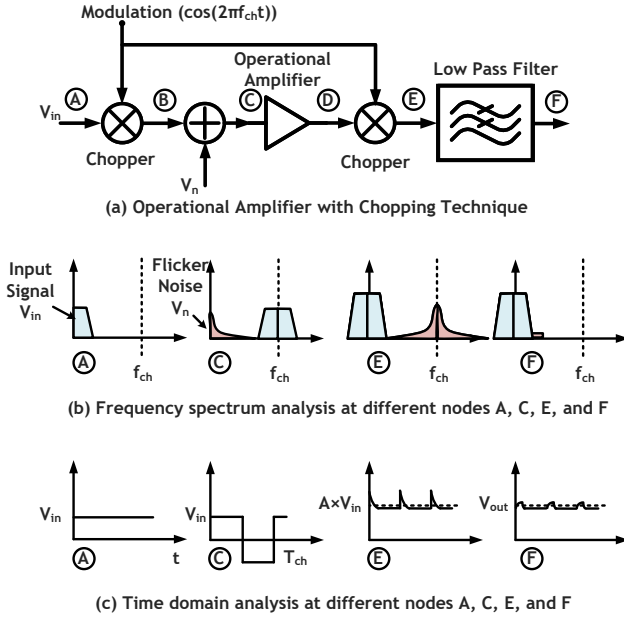


Fig. 4. The working principle of chopper amplifier: (a) an operational amplifier using chopping technique, (b) frequency spectrum analysis at different nodes and (c) time domain analysis at different nodes.

To achieve an optimal design trade-off between noise and power consumption in a chopper amplifier, we first need to understand the working principle and then investigate the influences of chopping frequency, the type of switches and its non-idealities on its noise performance. In this work, our study includes the optimization of f_{ch} , types of switches, and its associated parasitic capacitors. Table I details the non-ideal parameters used in evaluating the effectiveness of the chopper amplifier. The simulation results are presented in Fig. 5.

A. Working Principle

As shown in Fig. 4(a), an operational amplifier using the chopping technique (chopper amplifier) [9] requires two modulation choppers and a low-pass filter (LPF) circuits. Figs. 4(b)-(c) present the frequency spectrum and time domain analyses at different voltage nodes of the chopper amplifier. The input signal, V_{in} (node A) is modulated to the chopping frequency, f_{ch} . Flicker noise, V_n (node B) is introduced by the operational amplifier near the DC frequency range. To eliminate the flicker noise after amplification (node D), the second modulation chopper shifts the input signal back to its original frequency band (node E) while modulating V_n to f_{ch} . V_n is suppressed with the use of LPF filter (node F). To avoid aliasing of the noise and offset V_n , f_{ch} is chosen to be at least $2\times$ of the bandwidth of V_{in} [9].

B. Effects of Chopping Frequency

Fig. 5(a) illustrates the input noise power spectrum density (PSD) of chopper amplifier with NS switches and chopping frequency f_{ch} , ranging from 200 Hz to 50 kHz. To quantify and analyze the noise accurately, we calculate the total input-referred-noise based on the electrocardiogram's frequency bandwidth from 0.5 Hz to 150 Hz [22]. As shown

in Fig. 5(b), when the optimized chopper amplifier operates at 2 kHz chopping frequency, it can achieve $4.9\times$ reduction in the total input-referred-noise and decreases the flicker noise corner frequency by $1000\times$ (the flicker noise corner frequency is defined as the frequency where thermal and flicker noise equally contribute to the PSD [23]) compared to the amplifier without using chopping technique. However, considering the power-noise design trade-offs, we observed that the chopper amplifier is able to achieve the most energy-efficiency when its chopping frequency of 1 kHz f_{ch} . It can save up to 50% power consumption while increasing 20% noise compared to the chopping frequency of 2 kHz (lowest noise performance).

C. Effects of Parasitic Capacitors

To quantify the noise performance degradation of the chopper amplifier due to the effect of parasitic capacitance, we have evaluated the circuit using a chopping frequency of 2 kHz and varying NS switches' parasitic capacitance from 25 fF to 200 fF. Furthermore, we also evaluate how the mismatch (mismatch ratio) in parasitic capacitance from the complementary switches affects the noise performance. Figs. 5(c)-(d) present the simulation results of the PSD and the total input-referred-noise. We observed that the noise performance is nearly inversely proportional to the magnitude of parasitic capacitance and mismatch ratio. Under the most ideal situation (25 fF, no mismatch), the chopper amplifier can achieve $10\times$ flicker noise corner frequency reduction compared to that of the largest parasitic capacitance (200 fF, no mismatch). Moreover, with the same parasitic capacitance (200 fF) under 87.5% mismatch, the total input-referred-noise increases $2\times$ compared to that of without mismatch. Therefore, it is important to design the physical layout carefully while ensuring layout symmetry to lower down the parasitic capacitance of chopper switches and minimize the mismatch ratio.

D. Effects of Switches in Chopper Amplifier

To identify the most suitable switch for the chopper amplifier, we have evaluated the four types of switches (NS , $NS-D$, TG , BS) and the results (PSD and total input-referred-noise) are presented in Figs. 5(e)-(f). As shown in Fig. 5(e), the use of different type of switches has different degree of parasitic capacitance, affecting the flicker noise corner frequency. The chopper amplifier with BS achieves the lowest total input-referred-noise performance with 5% noise reduction compared with NS . Comparing with NS , BS achieves 5% noise reduction while TG and $NS-D$ cause extra 10% and 5% total input-referred-noise, respectively. Although $NS-D$ switch is known for its benefits in reducing the charge injection and clock feed-through effects, the additional parasitic capacitance from these dummy capacitance introduce mismatch between the switches, increasing the total input-referred-noise. Similar, the use of TG switches introduces more mismatch between the switches and it also demands an ideal complementary chopper clock signal, which is impractical in design. Therefore, we prefer to use BS to

TABLE I
THE NON-IDEAL PARAMETERS USED IN EVALUATING THE EFFECTIVE OF
CHOPPER AMPLIFIER.

Parameters	Range	Steps
f_{ch}	200 Hz ~50 kHz	200, 1k, 2k, 5k, 10k, 50kHz
C_{gs}	25~200 fF	25 fF (linear step size)
C_{gd}	25~200 fF	25 fF (linear step size)
C_{p1}	25~200 fF	25 fF (linear step size)
C_{p2}	25~200 fF	25 fF (linear step size)

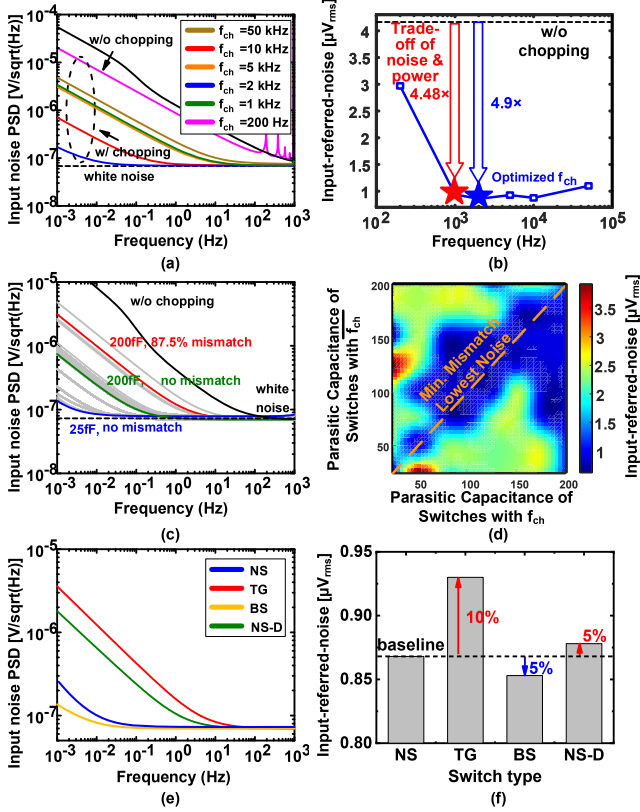


Fig. 5. The relationship between the non-idealities of chopper and noise suppress effects.

achieve a lower total input-referred-noise and higher signal output's linearity [24].

IV. CONCLUSIONS

In this work, through our analytical study with simulation results, we have seen that the choice of chopping frequency, the type of switches and its non-idealities affects its noise performance. To achieve the lowest total input-referred-noise of the amplifier, there is a need to (1) identify the most optimal chopping frequency, $f_{ch,optimized}$, (2) reduce the parasitic capacitors of switches and ensure proper symmetry in the physical layout, and (3) using BS switch to achieve a lower total-input-referred-noise and higher signal output's linearity.

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